

INTERNSHIP PROJECT REPORT

NEXUS ROBOTICS

AI Factory Management Dashboard

React + TypeScript

Three.js 3D Twin

Real-time Telemetry

01 PROJECT PROBLEM STATEMENT

The Problem

Modern industrial robotics facilities lack unified, real-time monitoring and control interfaces. Engineers operating multi-robot factory floors must juggle fragmented dashboards, disconnected telemetry streams, and reactive (not predictive) maintenance workflows — leading to costly downtime, missed anomalies, and slow incident response.



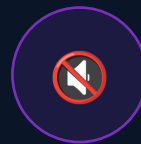
No Unified Control

Robot fleets managed through multiple disconnected tools with no single pane of glass.



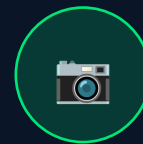
Reactive Maintenance

Failures discovered only after breakdown — no predictive wear or risk analytics.



No Acoustic AI

Mechanical anomalies invisible to operators; no ML-powered frequency diagnostics.



Manual Camera Monitoring

CV surveillance feeds unintegrated; no automated object detection or alerting.

NEXUS-ROBOTICS is a full-stack AI robotics management dashboard that consolidates real-time fleet telemetry, 3D digital twins, predictive maintenance, computer vision feeds, acoustic ML diagnostics, and an AI command assistant into a single futuristic SCADA-grade interface — built entirely client-side with no backend dependency.

Unified Fleet Dashboard

Single interface for all 6 robots with live KPIs, power draw, and thermal heatmaps.

3D Digital Twin (Three.js)

Real-time ABB IRB 6700 robot arm with manual joint control and auto telemetry sync.

Predictive Maintenance Engine

ML risk scoring from temperature + efficiency data; prevents failures before they occur.

Acoustic Anomaly Detection

32-bar frequency spectrum analyser with ML inference health monitoring.

AI Command Assistant

NLP-powered chatbot responds to natural language queries with live telemetry data.

Computer Vision Feeds

8-channel simulated CV camera surveillance with automated anomaly detection labels.

03 TECH STACK & TOOLS USED



React 18

UI Framework



TypeScript

Type Safety



Vite 6

Dev Server / Build



Tailwind CSS v4

Utility Styling



Three.js

3D Robot Twin



Recharts

Data Visualisation



Framer Motion

Animations



Lucide React

Icon Library



LocalStorage

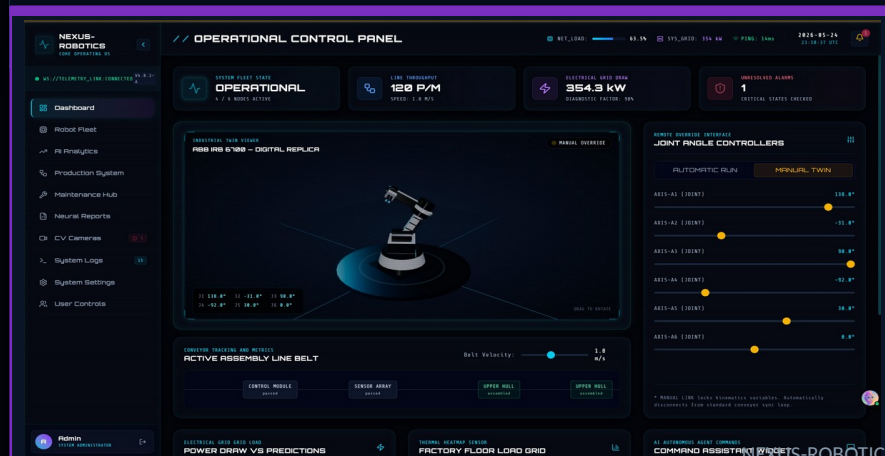
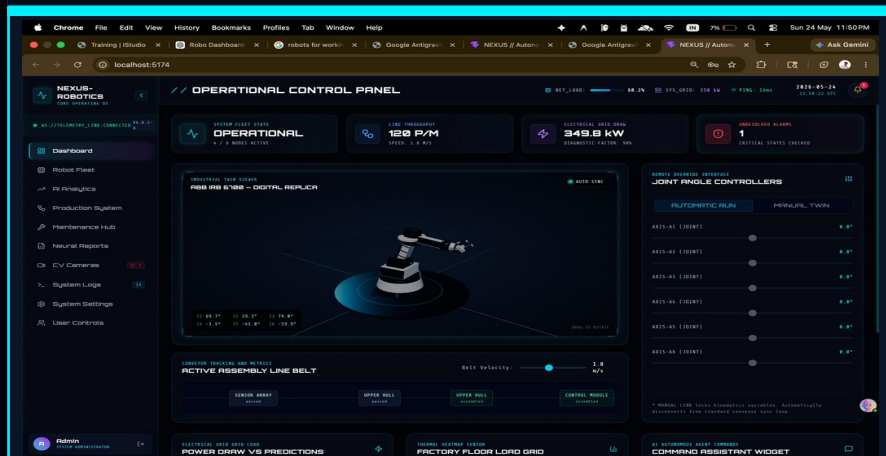
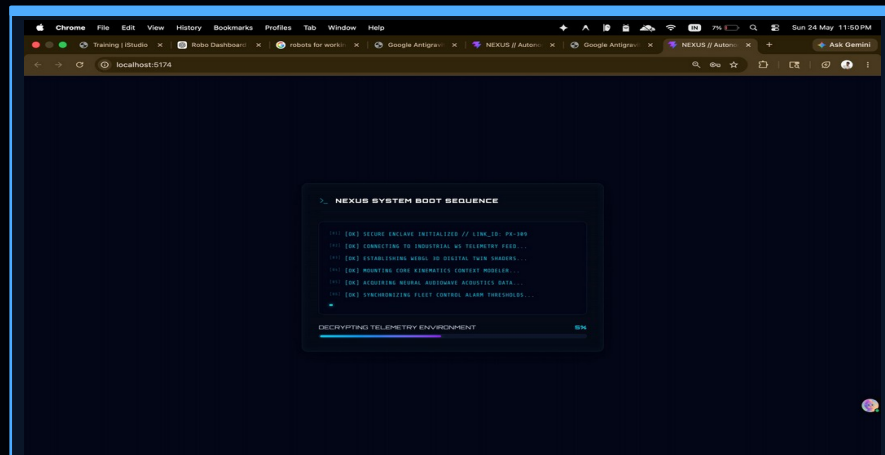
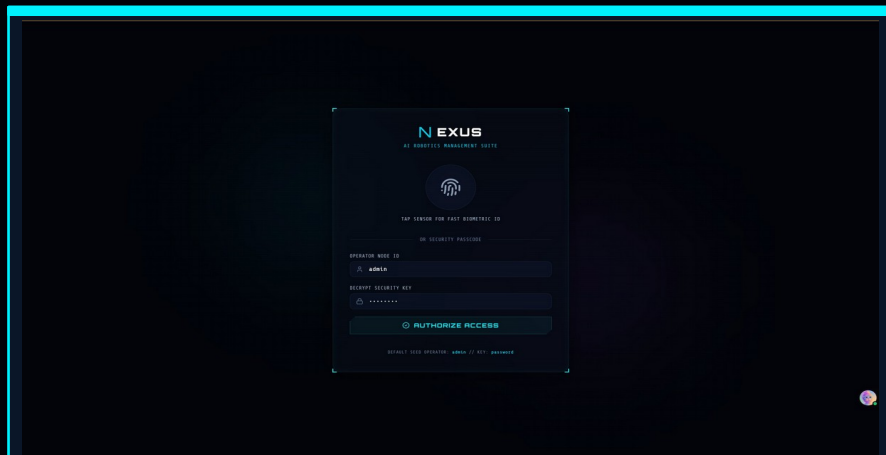
Log Persistence

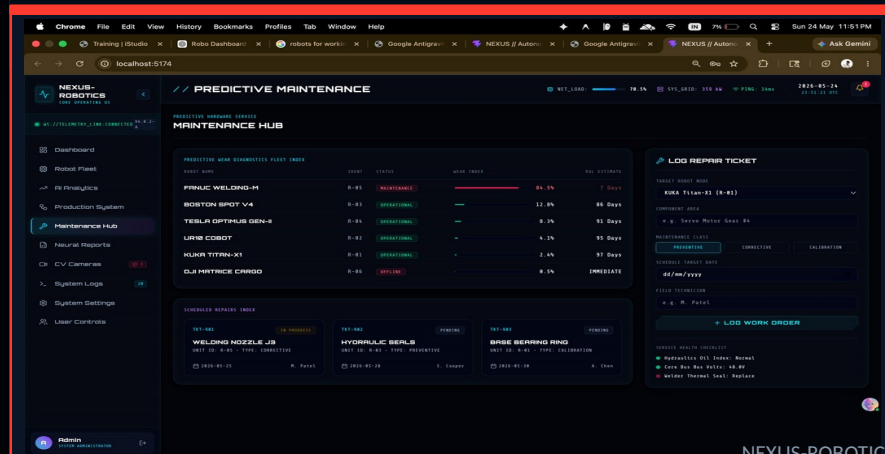
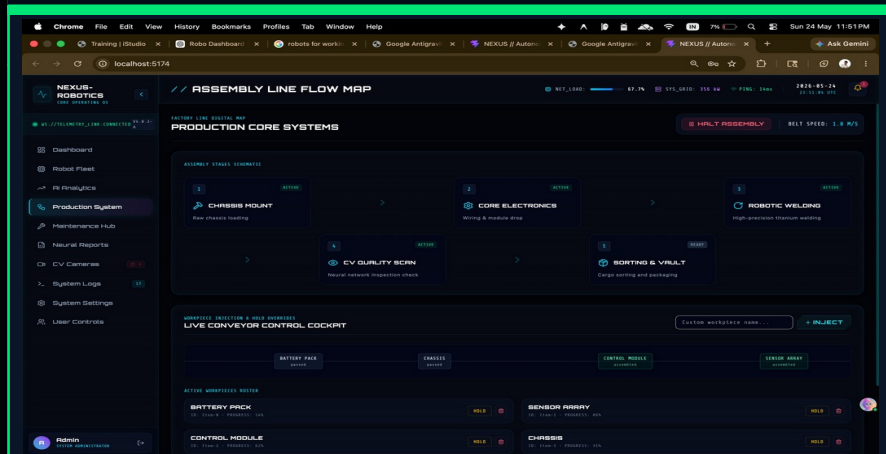
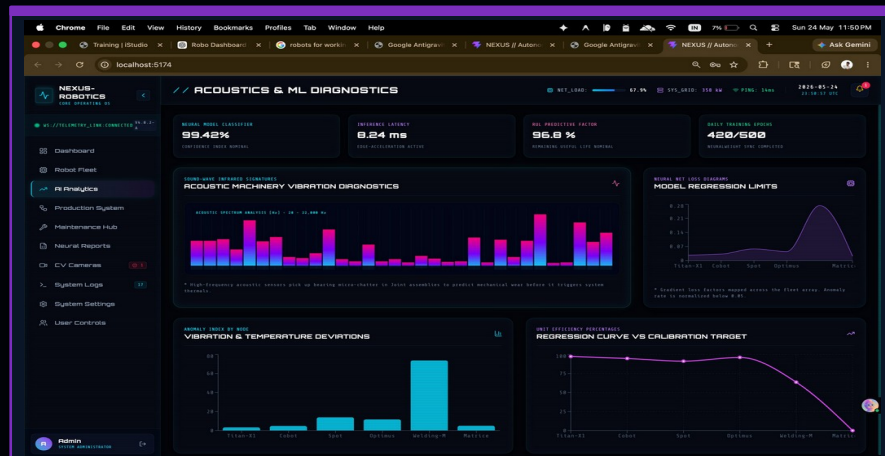
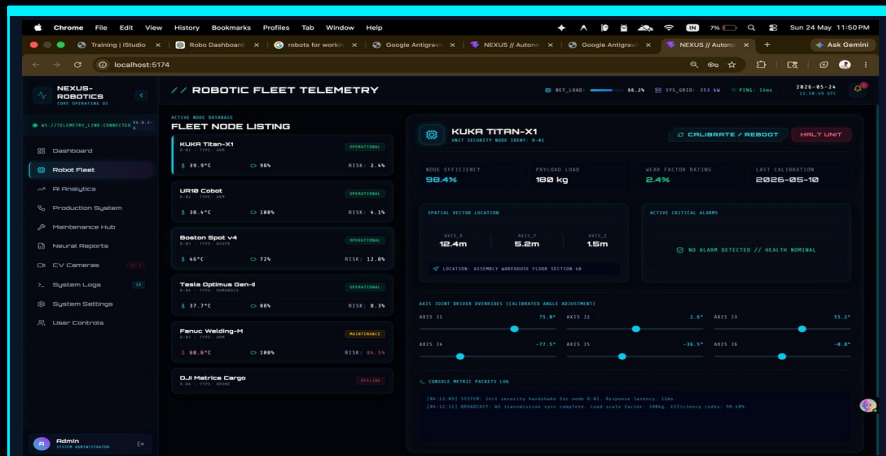


WebSocket (sim)

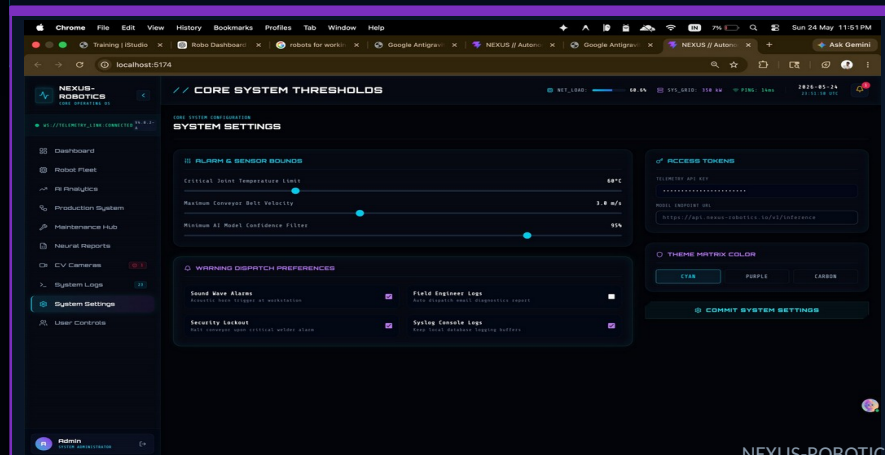
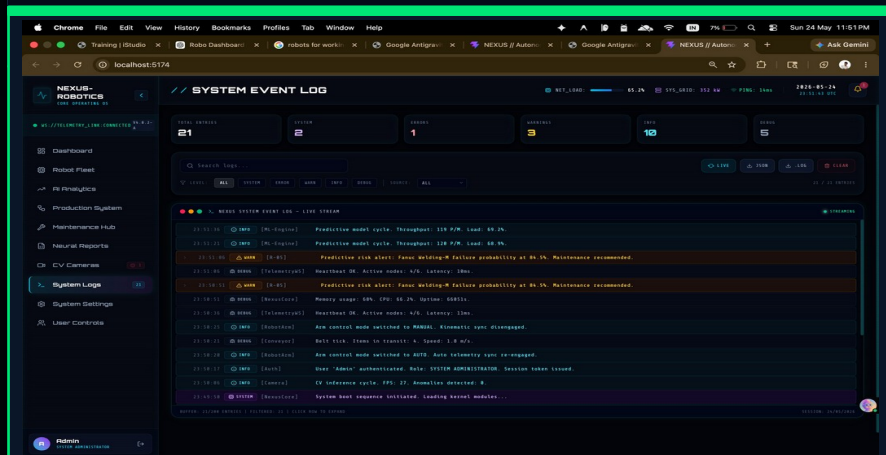
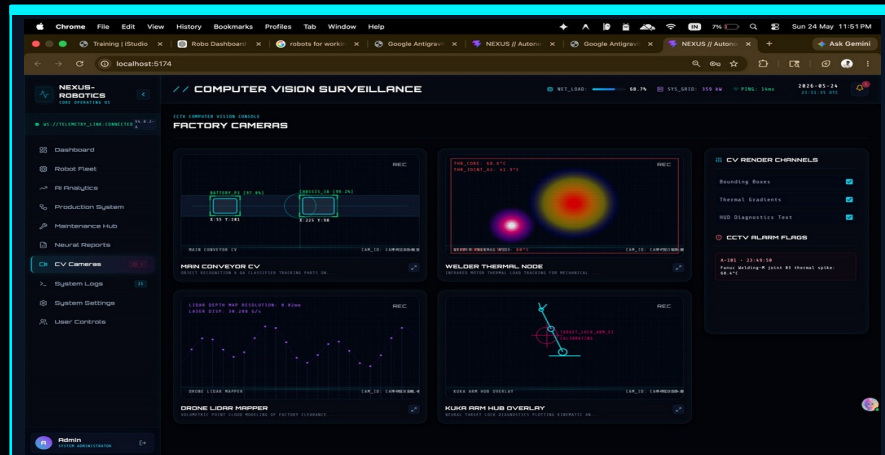
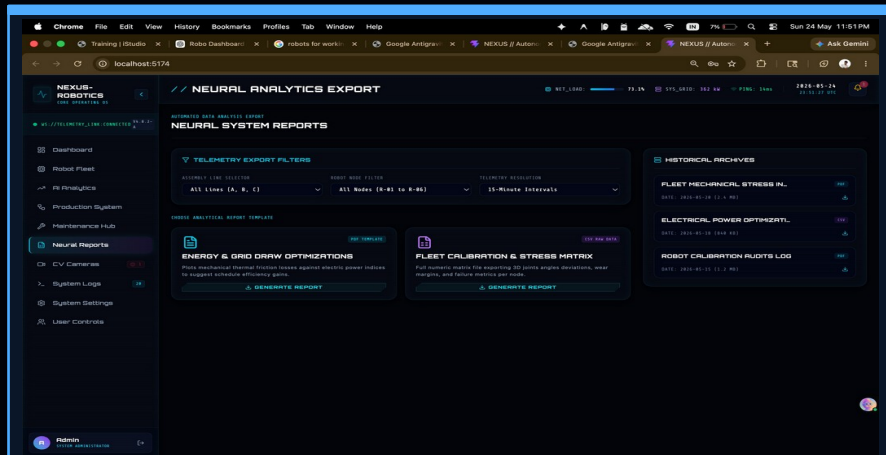
Live Telemetry

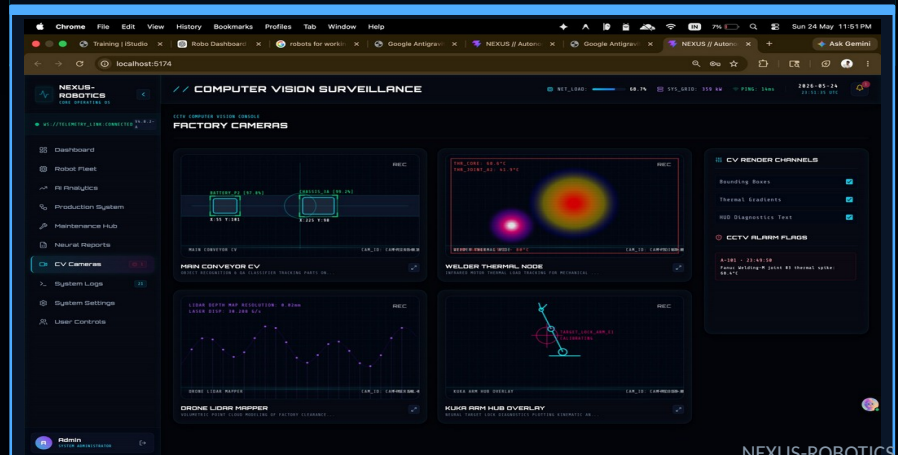
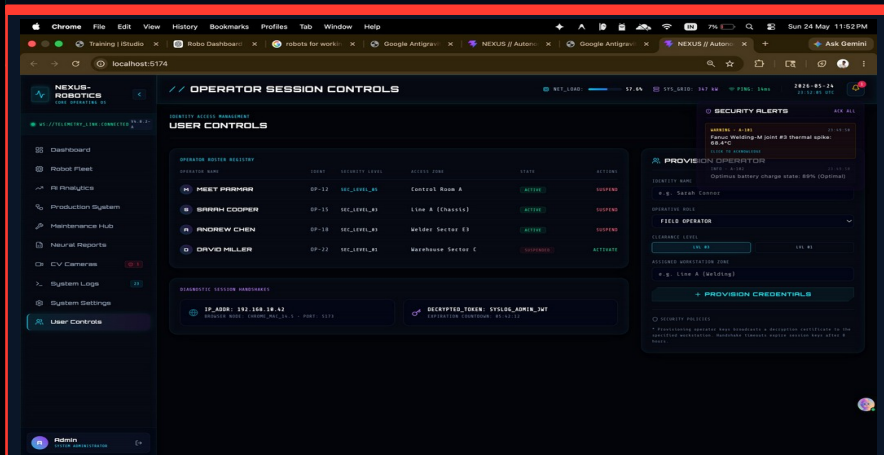
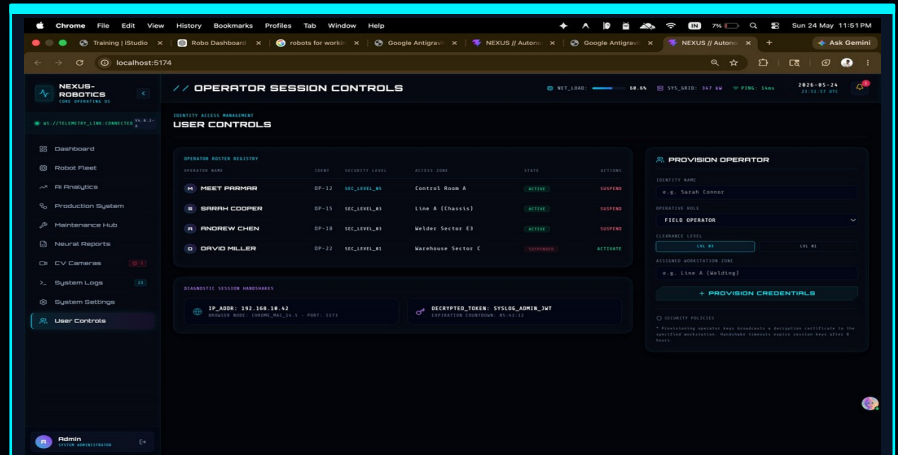
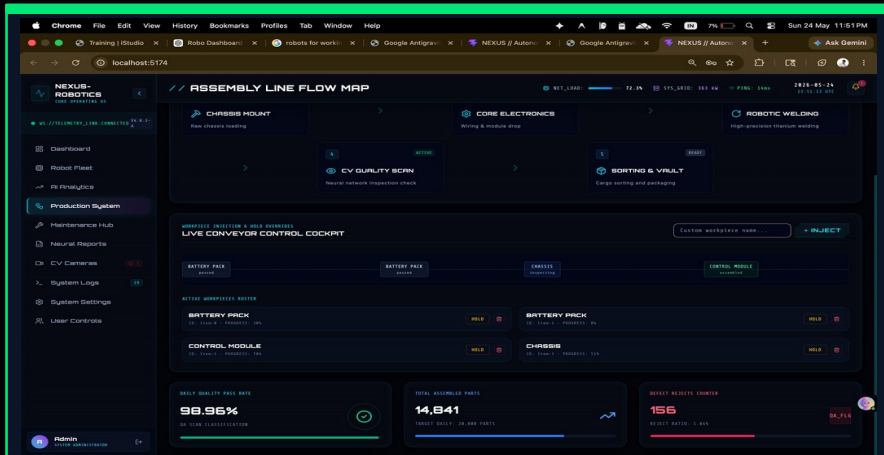
04 PROJECT SCREENSHOTS — LOGIN & DASHBOARD





06 PROJECT SCREENSHOTS — REPORTS & MONITORING





08 SYSTEM ARCHITECTURE

PRESENTATION LAYER

Dashboard · Fleet Monitor · AI Analytics · Production System · Maintenance · Reports · CV Cameras · System Logs



STATE / CONTEXT LAYER

TelemetryContext.tsx — Global state management, simulated WebSocket stream, 1.5s robot update loop, 15s log emitter



DATA / SIMULATION LAYER

Simulated robots R-01...R-06 · Conveyor items · Power draw · Acoustic data · Predictive ML scores · Auth session

✓ 100% client-side — No backend required. All telemetry, ML scores, log streams and CV feeds are simulated in the browser via TelemetryContext.tsx. Replace setInterval with a native WebSocket to connect to a real data source.

09 KEY OUTCOMES & PROJECT METRICS

10+

Dashboard Pages

6

Simulated Robots

8

CV Camera Feeds

1.5s

Telemetry Update Rate

WHAT WAS ACHIEVED

- Built a production-grade React + TypeScript SCADA interface with 10+ fully functional pages, zero backend dependencies.
- Integrated Three.js 3D digital twin of an ABB IRB 6700 robot arm with live telemetry sync and manual 6-axis joint control.
- Implemented ML-based predictive maintenance risk scoring, acoustic anomaly detection, and AI NLP command assistant.
- Designed a complete design system with glassmorphism panels, neon glow tokens, particle background, and CRT scanline overlay.
- Delivered log persistence via LocalStorage, CSV/JSON/PDF export, and simulated WebSocket stream — extensible to real data.

10 PROJECT REPORT SUMMARY

Project Overview

NEXUS-ROBOTICS is an ultra-premium AI robotics management dashboard designed to replicate enterprise SCADA/robotics OS solutions. The project was built using React 18 + TypeScript with Vite 6 as the build tool, using Tailwind CSS v4 for styling and Three.js for real-time 3D visualisation.

The system simulates a complete factory automation environment: 6 robotic units with live telemetry, an 8-channel CV surveillance system, an ML-driven predictive maintenance engine, acoustic anomaly detection, a production line conveyor manager, and a natural language AI command assistant — all running in the browser without a backend server.

Skills Demonstrated

- React component architecture with Context API for global state
- TypeScript interfaces and type-safe prop design
- Three.js 3D scene setup, lighting, and animation loops
- Recharts integration for live data visualisation
- Tailwind CSS v4 design system with custom CSS tokens
- Simulated real-time data streams and WebSocket pattern

Tools & Technologies

- React 18 + TypeScript + Vite 6
- Three.js for 3D digital twin rendering
- Recharts for area/radar/bar charts
- Tailwind CSS v4 + custom glassmorphism CSS
- Framer Motion for UI animations
- LocalStorage for log persistence & session data

THANK YOU

NEXUS — ROBOTICS

AI Factory Management Dashboard · Internship Project

React + TypeScript

Three.js 3D Twin

Predictive AI Engine

Built with React 18 · TypeScript · Vite 6 · Three.js · Tailwind CSS v4 — I Studio Project · Robotics Internship 2026

Demo Credentials

admin / any password · operator / any password